SU3plus: An Enhanced Swin-UNet for Synthesizing Vertically Integrated Liquid From Multiple Meteorological Satellite Data

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***Abstract*— Radar composite reflectivity is a crucial component of Earth observation data, playing a significant role in applications such as weather forecasting and climate disaster tracking. Due to the deployment challenges and limited coverage of meteorological radars, it is impossible to collect corresponding radar reflectivity in areas such as mountains and oceans. In such cases, using deep learning (DL) methods to reconstruct radar reflectivity from meteorological satellite data becomes an effective solution. However, Earth observation data is complex and exhibits strong long-range dependencies. Such data characteristics require the ability to model long-distance dependencies, making the Transformer more suitable for this specific data. With the ultimate goal of producing high reconstruction quality, we propose in this article a novel architecture, designated by SU3plus, based on the Swin Transformer with a multiscale feature fusion (MSFF) mechanism, while leveraging multiple channels of satellite data. Moreover, we define an appropriate loss function by combining the weighted versions of standard metrics, to take into account the data distribution imbalance and improve the reconstruction performance. Extensive experiments, carried out on the storm event imagery (SEVIR) storm dataset, confirm the effectiveness of the proposed approach compared to several state-of-the-art models.**

***Index Terms*— Data reconstruction, fusion, meteorological satellite data, neural network, radar composite reflectivity.**

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I. Introduction

METEOROLOGICAL satellites provide a holistic view of the earth system by using several input data, which are correlated and obtained at different spatial locations. Events such as storms, lightning, and hail, which are random and complex, affect human daily life [1]. In these situations, severe convective weather affects transportation, agriculture, forest fires, and floods. Accurate and comprehensive observation, monitoring, and prediction of Earth's variable meteorological information can help humans better avoid disasters or use natural resources [2], [3], [4], [5]. Meteorological radar data, with its accuracy, timeliness, and high spatiotemporal resolution, is one of the important supports for Earth system observation data [6], [7]. Meteorological radar uses electromagnetic waves to detect targets such as water droplets and ice crystals in the atmosphere and can measure the position, movement, shape, and characteristics of the targets [8]. The generation of vertically integrated liquid (VIL) data is based on the integration of radar echo signals, reflecting the size and density distribution of precipitation particles, and so, it is often used to represent the intensity of weather targets. However, due to the terrain limitations of radar deployment, special terrains such as mountains and oceans cannot collect the corresponding VIL. This limitation has become one of the main factors restricting the application of meteorological radar.

Compared to meteorological radar, geosynchronous meteorological satellites are not limited by terrain conditions and natural environments. They can monitor the Earth in all directions, have a wide coverage, and are suitable for observing systems that are large in scale and not short in duration [9]. Up to now, meteorological satellite detection technology has made many breakthroughs, with more detection channels and the ability to detect from multiple bands. Information such as infrared information, cloud top height, cloud top thickness, and water vapor information provide strong support for the meteorological industry. Meteorological radar data and meteorological satellite data reflect the same meteorological information for the same spatiotemporal position, and there is a strong correlation between them. This enables synthesizing meteorological radar data through meteorological satellite

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data, which is a new direction for inverting meteorological radar data [10], [11], [12], [13].

In this context, DL is a powerful tool for solving data reconstruction problems, which is widely used in the field of meteorology [14], [15], [16], [17], [18], [19], [20]. DL models are not designed for specific physical rules but are trained using a large amount of meteorological satellite data and meteorological radar data. By learning from a large amount of data, DL models will show strong modeling capabilities, refine the physical rules in the meteorological system, and synthesize more accurate VIL. In this respect, recent research efforts, based on DL approaches, have been developed and have shown promising results [21]. These solutions mainly rely on convolutional neural networks (CNNs) and recurrent neural networks (RNNs) such as [22], [23], [24], and [25]. These two architectures impose inductive bias in time and space and have been proved to help capture patterns in time and space. However, the meteorological information of each pixel of VIL is related to global information; and small-scale severe convective weather is the focus of attention, which poses a greater challenge to the model's long-distance dependence ability [26]. Although CNNs and RNNs can pay attention to the relevance of local information, they cannot model global information [27]. To address this limitation, a more efficient architecture, such as the Transformer, can be used because of its ability to better capture data correlation through attention mechanisms.

This makes the Transformer more adapted to model long-distance dependencies and global information in meteorological radar data. Currently, new meteorological prediction networks based on the Transformer architecture have been proposed. For instance, Earthformer [27] and Rainformer [28] have been developed for rainfall prediction and climate reporting tasks, respectively, showing the potential of Transformers for atmospheric modeling purposes. However, the design strategy and the objective of the latter models differ significantly from our target radar reflectivity reconstruction task. In this respect, different Transformer-based models have been developed. The latter will be described in Section II.

The objective of this article is to design a novel architecture for the reconstruction of VIL data. Specifically, we first propose to leverage lightning information (LGHT) in addition to satellite infrared channels (IR-069 and IR-107), which yields a more accurate reconstruction. Moreover, we develop a Transformer network architecture called SU3plus, based on Swin-UNet [29], which aims to fully exploit the powerful long-distance dependencies in the meteorological data. To this end, a multiscale feature fusion (MSFF) block is built to enable the recovery of both global and local information during the reconstruction process. This design integrates low-level details and high-level semantics from feature maps at different resolutions, which will maximize the use of multiscale feature maps of the input data. Furthermore, we define an appropriate weighted loss function, which combines a mean-square-error (mse) metric and a structural similarity index (SSIM), for the training of the proposed architecture. Thus, SU3plus can grasp the overall spatial form of the Earth's meteorological system and more accurately reconstruct the high-threshold details of the VIL data. Extensive experiments, performed on the SEVIR dataset [30] (some samples of which are illustrated in Fig. 1) show the relevance of different design choices and confirm the potential of the designed VIL synthesis architecture.

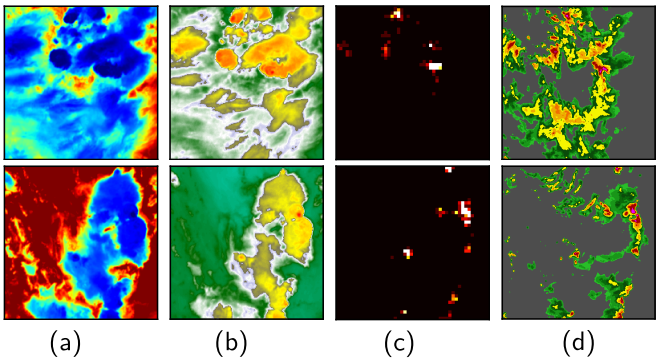


Fig. 1. Examples of images in the SEVIR dataset. (a) IR-107 is a satellite cloud image. (b) IR-069 is a satellite infrared image. (c) LGHT is a visualization of lightning.(d) True radar VIL image. (a) IR-107. (b) IR-069. (c) LGHT. (d) VIL.

The remainder of this article is organized as follows. Section II gives an overview of the related methods developed for meteorological radar data reconstruction. Section III describes the proposed model architecture. Section IV presents the experimental results based on objective and subjective evaluation. Ablation studies are also performed to demonstrate the effectiveness of our design methodology. Finally, Section V concludes this article and addresses some perspectives.

II. Related Work

Different research efforts, based on DL approaches, have recently been made for VIL reconstruction purposes. The developed methods focus on some key aspects including input image types, network structures, and loss functions to gradually improve model performance. For instance, these methods employ CNN architectures, where most of them correspond to UNet and its variants.

A. DL Methods for Radar VIL Data Reconstruction

In recent years, DL methods have attracted particular attention in the literature. For instance, Hilburn et al. [31] have designed a CNN network that uses the UNet to convert GOES-R radiation and lightning into synthetic radar reflectivity fields. The authors believe that geostationary lightning observations have unique value because they can accurately locate strong radar echo locations. Veillette et al. [32] used CNNs to fuse satellite radiation data, ground meteorological station data, and lightning data to generate synthetic VIL. Moreover, Veillette et al. [30] have created a public dataset, referred to as SEVIR, which is specifically designed for DL applications in radar and satellite meteorology. Duan et al. [33] trained a CNN model using five infrared bands of the Himawari-8 satellite to convert radiation data into composite radar reflectivity factors. They observed that there is a serious data imbalance in the radar composite reflectivity data and proposed a training domain with a patch selection approach to improve the synthesis performance. Recently, Yang et al. [34]

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and Cui et al. [35] used the attention UNet, which is trained using inputs with and without visible light separately. They show the effectiveness of visible light data in improving model performance. It has also been shown that the attention UNet performs better than the UNet. Wang [36] proposed an enhanced UNet-based architecture to reconstruct radar images by integrating multiresolution data inputs. Xiang et al. [37] applied the Swin-Unet to process FY-4B satellite imagery for meteorological applications. Yin et al. [38] adopted the UNet3+ architecture to generate and detect cloud and snow features from Landsat-8 satellite images. Liu et al. [39] leveraged the TransUnet to reconstruct radar reflectivity from FY-4A satellite observations. Lin et al. [22] proposed a method combining convolution and MLP, which uses encrypted meteorological satellite remote sensing data to improve the accuracy and security of radar reflectivity synthesis. Gavahi et al. [40] used the 3D-CNN to perform rainfall prediction and climate reporting tasks. However, all of these methods are based on the CNN architecture and do not make full use of effective data. In this work, we replace the basic CNN module with a Swin Transformer-based architecture and leverage multiple channels of satellite data, yielding better reconstruction performance. Si et al. [41] used the FR-CNN to reconstruct the radar reflectivity.

B. Effectiveness of the Swin Transformer in Computer Vision

The Swin Transformer is a hierarchical Visual Transformer that uses shifted windows to enhance the receptive field and self-attention efficiency while maintaining computational and memory scalability [42]. Thus, it can exploit image inputs at different scales and resolutions and demonstrate strong generalization capabilities. For instance, the Swin Transformer has been adopted in the popular Swin-UNet architecture, which has recently been designed for medical image segmentation [29]. It draws on the U-shaped encoder–decoder architecture and skips connections of the UNet [43]. Recent advances in image processing, and more specifically in remote sensing context, have leveraged the Swin Transformer, as demonstrated by Tu et al. [44] who combined it with the CNN in a generative adversarial network for super-resolution, and Zhou et al. [45] who embedded it into a dual-stream network for semantic segmentation. He et al. [46] proposed an ST-Unet for remote sensing image semantic segmentation. Zhao et al. [47] proposed a DDF-former for ocean wind speed estimation. However, to the best of our knowledge, such a Transformer architecture has not yet been investigated for radar data reconstruction.

III. Methodology

This article aims to design an efficient approach for the reconstruction of VIL data. The proposed methodology relies on the effective fusion of multichannel satellite data to improve the accuracy and spatial resolution of radar reflectivity. Knowing that there are relatively few works on the reconstruction of radar VIL data using deep CNNs and more specifically UNet-based architectures, this article attempts to: 1) leverage satellite infrared channel images (IR-069 and IR-107) as well as satellite lightning data (LGHT) and 2) exploit the Transformer model to design the network architecture while incorporating an efficient fusion block.

A. SU3plus Architecture

To fully exploit the characteristics of long-distance dependence and capture the internal physical rules of high-resolution meteorological data, SU3plus mainly enhances the network's long-distance dependence and global modeling ability through the following three mechanisms: a U-shaped hierarchical structure, an MSFF block, and a Swin Transformer module.

1) Overall Architecture: As shown in Fig. 2, the proposed SU3plus network is inspired by the U-shaped encoder–decoder structure, composed of an encoder, a bottleneck layer, and a decoder. Its design relies on the following main steps. First, the infrared channels and the lightning information are concatenated and processed. Thus, Patch Extract and Patch Embedding are used to divide the merged image into nonoverlapping blocks and send them to the encoder. In the encoder part, three groups of Swin Transformer modules of different scales are used to extract the context features of the image blocks and perform downsampling. In the bottleneck layer, a group of Swin Transformer modules with a larger window size is used to enhance global information interaction. In the decoder part, we designed the MSFF block and combined it with the Swin Transformer module to restore both the global and local features of the image. Finally, the Patch Expanding step is employed to upscale the feature vectors and produce the final output reconstructed image.

2) MSFF Mechanism: We designed an MSFF block to address the imbalance in restoring high-threshold and low-threshold features. The structure of the proposed MSFF block is shown in Fig. 3. In meteorological radar data, both high-threshold and low-threshold information are equally important. However, because high-threshold regions are relatively small in the image, traditional methods with skip connections struggle to balance these features, leading to insufficient reconstruction of high-threshold information. Our design effectively integrates both types of features.

More precisely, the MSFF block consists of a feature adjustment layer and a dense layer. The feature adjustment layer adjusts the dimensions of the received feature maps, while the dense layer fuses features from different scales. Thus, the MSFF block allows the decoder to receive information from encoder feature maps at arbitrary scales, rather than being restricted to feature maps of the same scale. Moreover, it transmits the fused multiscale features to the next decoder layers as additional feature cues. Note that for high-resolution data, upsampling and downsampling often result in the loss of crucial semantic information. The fusion of feature maps from different levels effectively reduces this loss of information. To further minimize the loss caused by pooling and transposed convolution operations, the MSFF block employs Patch Merging and Patch Expanding to adjust feature map scales.

Therefore, this design approach enables better exploitation of high-level semantic information and low-level detail information in the network, facilitating the integration of features

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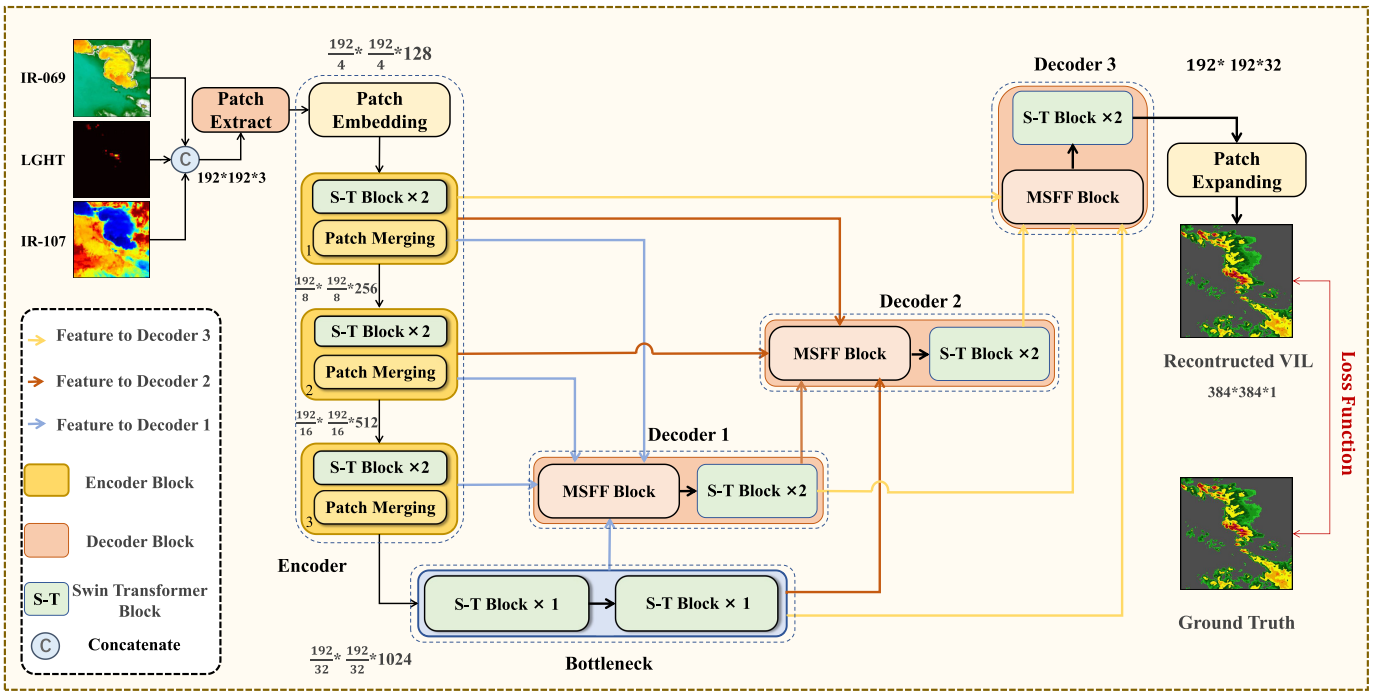


Fig. 2. SU3plus mainly strengthens the long-range dependence performance and global modeling capability of the network through the following three aspects: a U-shaped hierarchical structure, a Swin Transformer module, and an MSFF block. This is to fully utilize the characteristics of long-range dependence to capture the physical rules within high-resolution meteorological data. We can see that by concatenating IR-107, IR-069, and LGHT as input, and computing the loss between the model's output and the ground truth, the overall structure is an end-to-end architecture.

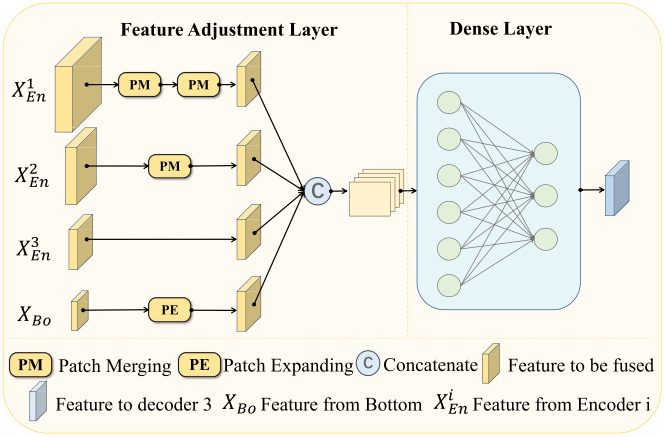


Fig. 3. Structure of the MSFF block. The MSFF module consists of a feature adjustment layer and a dense fusion layer. The feature adjustment layer adaptively calibrates the dimensions of the input feature maps and is composed of multiple patch merging or patch expanding layers. The dense fusion layer then integrates features across scales. This design allows the decoder to break free from the same-scale skip-connection limitation of the traditional UNet architecture and dynamically ingest multiscale features from any encoder level.

across different layers and improving reconstruction accuracy. In summary, the fused features of the MSFF blocks employed at the three decoder layers can be expressed as

$$\begin{cases} F_{\mathrm{De}}^{1} = \mathrm{Cat}(E(X_{\mathrm{Bo}})X_{\mathrm{En}}^{3}, M(X_{\mathrm{En}}^{2})M(X_{\mathrm{En}}^{1})) \\ \hat{X}_{\mathrm{De}}^{1} = D(F_{\mathrm{De}}^{1}) \end{cases} \tag{1}$$

$$\begin{cases} F_{\mathrm{De}}^{2} = \mathrm{Cat}(E(X_{\mathrm{Bo}})E(X_{\mathrm{De}}^{1}), X_{\mathrm{En}}^{2})M(X_{\mathrm{En}}^{1})) \\ \hat{X}_{\mathrm{De}}^{2} = D(F_{\mathrm{De}}^{2}) \end{cases} \tag{2}$$

$$\begin{cases} F_{\mathrm{De}}^{3} = \mathrm{Cat}(E(X_{\mathrm{Bo}}), E(X_{\mathrm{De}}^{3}), E(X_{\mathrm{De}}^{2}), X_{\mathrm{En}}^{1}) \\ \hat{X}_{\mathrm{De}}^{3} = D(\mathrm{F}_{\mathrm{De}}^{3}) \end{cases} \tag{3}$$

where X_{En}^{i} represents the feature map produced from the ith encoder layer, X_{Bo} denotes the feature map generated from the bottom layer, and X_{De}^{i} refers to the feature map produced from the ith decoder layer. M stands for the Patch Merging operation, E indicates the Patch Expanding operation, Cat represents the feature concatenation operation, F_{de}^{i} represents the features to be fused at the ith decoder layer, D signifies the fully connected operation, and $\hat{X}_{\mathrm{De}}^{i}$ represents the input of the ith decoder layer.

3) Integration of the Swin Transformer: The integrated Swin Transformer introduces two types of window-based self-attention mechanisms to achieve long-distance dependence. The first is the window-based multihead self-attention (W-MSA), which divides the input image into nonoverlapping windows, and then performs self-attention calculation in each window. The second is the shifted window-based multihead self-attention (SW-MSA), which cyclically shifts the input image, and then divides it into windows. Such a mechanism allows the model to realize cross-window information exchange. Thus, using the Swin Transformer module in the reconstruction task can better capture the global context information of the image and improve global perception capabilities.

B. Learning Approach

1) Weighted Loss Functions: To learn the model parameters, the predicted VIL image is compared to the ground truth based on a given loss function $\mathcal{L}$. The latter aims to combine

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two weighted versions of the mse and SSIM criteria. It is expressed as

$$\mathcal{L} = \mathcal{L}_{\mathrm{w-MSE}} + \mathcal{L}_{\mathrm{w-SSIM}}. \tag{4}$$

The weighted mse and SSIM loss functions $\mathcal{L}_{\mathrm{W-MSE}}$ and $\mathcal{L}_{\mathrm{W-SSIM}}$ are computed as follows:

$$\mathcal{L}_{\mathrm{W-MSE}} = \frac{1}{N}\sum_{i=1}^{N} W_i \mathrm{MSE}\left(y_i, \hat{y}_i\right) \tag{5}$$

$$\mathcal{L}_{\mathrm{W-SSIM}} = \frac{\sum_{i=1}^{N} W_i \left(1 - \mathrm{SSIM}\left(y_i, \hat{y}_i\right)\right)}{\sum_{i=1}^{N} W_i} \tag{6}$$

where N is the number of samples, y_i is the ground-truth image, $\hat{y}_i$ is the predicted image, $\mathrm{mse}(y_i, \hat{y}_i)$ and $\mathrm{SSIM}(y_i, \hat{y}_i)$ are the mse and SSIM criteria evaluated between y_i and $\hat{y}_i$, respectively, and W_i is a given positive weight associated with the image y_i. The motivation behind this weighting parameter and its setting are described in what follows.

2) Data Distribution and Weights Setting: Due to the overly large spatial scale of the Earth system, the range of severe convective weather is very small compared to the entire observation range. For instance, in the SEVIR dataset, the proportion of high-threshold radar reflectivity data is too small, and the dataset has a serious data imbalance problem. This data imbalance problem is determined by the nature of the Earth's meteorological system itself. In other datasets, such as the dataset collected by the Himawari-8 satellite used by [33], this problem also exists. As shown in Fig. 4, in the SEVIR dataset, the 0–16 range of radar data almost occupies 75% of the total data volume, 16–133 occupies 21%, and the remaining high-threshold areas only account for 4% of the total.

To take into account the data distribution imbalance, we proposed to weigh the standard mse and SSIM terms. Each data sample y_i is multiplied by the weight of its interval W_i, which is given by

$$W_i = \frac{1}{P(y_i)} \tag{7}$$

where $P(y_i)$ represents the probability of the interval containing y_i in the dataset. Note that setting the weights to the inverse of the probability $P(y_i)$ allows addressing long-tail distributions and avoiding instability in training. For instance, similar frequency-aware loss weighting has often been used in semantic segmentation and object detection tasks in the presence of class imbalance. Some related works addressing similar weighting strategies and an analysis of different weights will be presented in Section IV-E3.

The specific weight values W are shown in Table I. We believe that the proportion of weather phenomena with varying intensities within the Earth system remains constant over long time periods. Therefore, we can consider the statistical results as a universal prior (applicable across the Earth system), which can be incorporated into the algorithm.

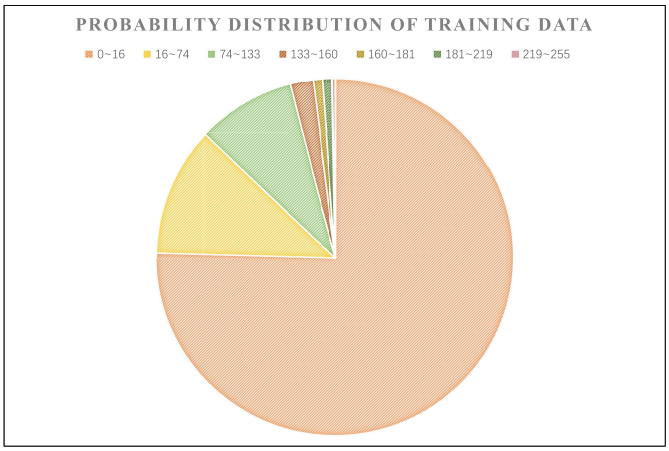


Fig. 4. Probability distribution of the meteorological data. The 0–16 range of radar data almost occupies 75% of the total data volume, 16–133 occupies 21%, and the remaining high-threshold areas only account for 4% of the total.

TABLE I
WEIGHT VALUES W

Interval	0-16	16-74	74-133	133-160	160-181	181-219	> 219
Weight	1.3363	8.4795	10.7041	45.0041	117.9627	133.3007	476.5719

IV. EXPERIMENTS AND RESULTS

A. Experimental Settings

Our simulations are carried out on a single NVIDIA V100 GPU, which has 32 GB of video memory and 48 GB of memory. During training, because of the huge amount of data, we use batch mode for learning. The input data is a tensor stacked on the channel dimension of three types of satellite images, and the shape of each group of inputs is $192 \times 192 \times 3$. For the SU3plus network, the learning rate is set to 0.0001. The UNet used in previous tasks is used as a control, and for the UNet network, the learning rate for training is set to 0.001.

B. Dataset

The public dataset storm event imagery (SEVIR) has been used for model training and testing. This dataset is a collection of meteorological images captured by satellites and radars, which are aligned in both time and space. It is constructed as a benchmark dataset to support algorithm development in meteorology. The dataset consists of two parts: directory and data. The directory part is a CSV file that contains metadata for each event, such as event ID, sensor type, time range, and spatial range; the data part is a set of HDF5 files. The SEVIR dataset includes five types of images and two types of events. The five types of images come from the 0.6-μm visible light channel of GOES-16, the 6.9- and 10.7-μm infrared channels of GOES-16, the radar VIL image of NEXRAD, and the geosynchronous lightning imager of GEOS-16. The two types of events are random events and storm events.

The VIL images in SEVIR are stored as integers in the range 0–254 and use the value 255 in regions where data is missing. It is often convenient to work directly with these encoded units. To convert these into units of kg/m^2, which are the true units of VIL, the following rule is applied:

$$f(X) = \begin{cases} 0, & \text{if } X \le 5 \\ \dfrac{X-2}{90.66}, & \text{if } 5 < X \le 18 \\ \exp\left(\dfrac{X-83.9}{38.9}\right), & \text{if } 18 < X \le 254 \end{cases} \tag{8}$$

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where X is the integer value stored in the HDF file. This nonlinear scaling rule was developed to make better use of the range 0–254 for storing VIL. The reason for this is that VIL data exhibits histograms that are quite skewed, and the nonlinear encoding provides better precision and lessens the skewness.

In practical applications, considering that radar images are often used for rainfall prediction, we use storm events in the radar data reconstruction model to enhance performance while reducing training time. Two infrared channel images and lightning data are used as inputs, and the radar composite reflectivity image is used as a label. The training set, validation set, and test set are divided in chronological order, with a data volume ratio of 70656:15520:24192. The training set consists of 70 656 groups of data from the start date to January 1, 2019, in the SEVIR dataset. The validation set consists of 15 520 groups of data from January 1, 2019, to June 1, 2019, in the SEVIR dataset. The test set consists of 24 192 groups of data from June 1, 2019, to the end date in the SEVIR dataset.

In addition, considering that in meteorological data arranged in chronological order, the similarity of data at adjacent times is too high, which may have a negative impact on the training of the synthetic meteorological radar model. Therefore, after dividing the training set, validation set, and test set, we randomly shuffle the data within each set.

Finally, we used the storm data index of the SEVIR dataset as training data. We used two infrared channel images (IR-069 and IR-107) and lightning data (LGHT) as inputs and true VIL as labels. Among them, the infrared channel can directly reflect water vapor information and lightning data can directly reflect information about small-scale severe convective weather. Since visible light information is missing at night, it is not used as a training input.

C. Evaluation Metrics

Different radar echo intensities represent various convective climate intensities. The threshold interval of the SEVIR radar data used in this article is [0, 255]. We set nodes at 16, 74, 133, 181, and 219, respectively. This results in six groups of radar data, where the synthesis effect is evaluated in each group.

From a classification point of view, we use the critical success index (CSI) as the main evaluation metric. The CSI is an effective evaluation metric suitable for the classification of imbalanced datasets and specific fields (such as meteorology). It provides a comprehensive evaluation of model performance by considering true positives (TPs), false positives (FPs), and false negatives (FNs)

$$\mathrm{CSI} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{9}$$

Compared with metrics such as precision and recall, CSI can better reflect the model's performance when dealing with minority classes. The closer the CSI is to 1, the more effective the prediction. In addition, from an image reconstruction point of view, we use the mean absolute error (MAE), mse, and SSIM metrics.

D. Quantitative and Qualitative Evaluations

As mentioned in Section II, most state-of-the-art methods rely on CNN and RNN methods [30], [33], while some recent methods have demonstrated the potential of models similar to the UNet and Transformer. Based on the previous works, we propose to compare our architecture with DL models that have been newly applied to meteorological radar image tasks. For these models, we trained them on the same dataset with weighted training, and the resulting quantitative metrics are shown in Table II.

From the quantitative results, it can be seen that our proposed SU3plus achieved the best results in mse, MAE, SSIM, and average CSI. Specifically, the average CSI reached 0.3862, yielding a gain of about 0.0710 compared to the second-best method TransUNet [39]. In the low-threshold range, our model yields similar performance to the Swin Transformer model [45]. In the mid-to-high threshold ranges, our model achieves higher gain and leads to the best results. The obtained results show that, for the state-of-the-art models, and in the absence of explicit guidance for learning in the high-threshold region, the reconstruction accuracy for low-threshold regions is generally good, but the accuracy in the mid-to-high threshold regions decreases. Some models, such as the Swin Transformer and UNet3+ [38], have near-zero synthesis success scores for threshold regions above 219. Most importantly, the proposed MSFF block achieves the expected effect, balancing features across scales, guiding the model to focus on image areas representing high-threshold regions, while maintaining reconstruction accuracy in the low-threshold regions.

From a qualitative analysis of the visualization results shown in Fig. 5, we selected two sets of meteorological data images under strong convective weather to compare the performance of each model. It can be seen that, for the pink areas representing the center of strong convective weather, only the proposed SU3plus model successfully reconstructed them. The other models only synthesized a small amount of red areas, representing severe weather phenomena. For the low-threshold range, models based on TransUNet and Swin-UNet [37] reconstructed regions that were not originally present, as shown in Fig. 5(f) and (i). We believe that these two models did not learn the physical weather information from IR-107 [Fig. 1(a)] and IR-069 [Fig. 1(b)], directly reconstructing cloud-covered areas as rainfall regions.

To further validate the generalizability of the different models, we divided the test dataset into four seasonal subsets based on the temporal scale and conducted quantitative evaluations of different models in each season. The results are presented in Table III. It can be clearly seen that SU3plus consistently outperformed other models across spring, summer, autumn, and winter in terms of SSIM, mse, and MAE metrics while maintaining higher accuracy across multiple CSI thresholds. In particular, in summer and autumn, SU3plus achieved average CSI scores of 0.4220 and 0.4219, respectively, significantly outperforming other methods and demonstrating its strong generalizability under varying climatic conditions. In contrast, other models show limited performance in winter, with greater fluctuations in metrics, indicating insufficient adaptation to seasonal variations. This experiment

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TABLE II
QUANTITATIVE EVALUATION OF THE PROPOSED SU3PLUS APPROACH AND STATE-OF-THE-ART MODELS

Network	UNet	Att-UNet	Swin-UNet	UNet3+	Swin-Transformer	TransUNet	SU3plus
Source	[36]	[35]	[37]	[38]	[45]	[39]	Ours
Metric	value						
SSIM↑	0.6511	0.7379	0.6360	0.7314	0.7311	0.6784	**0.7545**
MSE↓	1369.3506	896.3457	1325.5111	857.6481	945.292	906.3427	**711.741**
MAE↓	20.4267	13.6759	21.0239	13.6226	13.3005	16.5526	**12.2806**
CSI16↑	0.5290	0.6711	0.5081	0.6703	0.6743	0.5505	**0.6887**
CSI74↑	0.4174	0.5017	0.4162	0.5070	0.5085	0.4700	**0.5220**
CSI133↑	0.2993	0.2780	0.3379	0.3002	0.3054	0.3443	**0.3551**
CSI160↑	0.3316	0.1896	0.3346	0.2329	0.1691	0.3168	**0.3502**
CSI181↑	0.2325	0.0687	0.2178	0.1151	0.0327	0.2019	**0.3059**
CSI219↑	0.0221	0.0016	0.0048	0.0019	0.0001	0.0078	**0.0950**
CSI-avg↑	0.3053	0.2851	0.3032	0.3045	0.2817	0.3152	**0.3862**

demonstrates the robustness and stability of SU3plus over different temporal scales.

E. Ablation Studies

1) Effectiveness of Some Design Choices: First, we analyzed the contributions of incorporating the proposed MSFF block, weighting the loss terms, and leveraging the lightning information. In this respect, ten runs have been performed as shown in Table IV.

Thus, by comparing 1 with 3 and 6 with 8, we can find that without the MSFF mechanism, the CSI index decreased by 0.0062 and 0.0239, respectively. According to the various threshold intervals, it can also be seen that at medium and high thresholds, models using MSFF illustrate higher CSI gains. Therefore, we can conclude that MSFF effectively retains structural details and improves the synthesis accuracy of the model for positive examples.

Moreover, by comparing experiments 9 and 10, it can be observed that the absence of lightning information causes a significant decrease in the CSI in the mid-to-high threshold ranges: the CSI for low thresholds (0–16 and 16–74) decreases by 0.0330 and 0.0838, respectively, showing a relatively small impact; the CSI for mid-thresholds (74–133 and 133–160) drops substantially by 0.2351 and 0.3120, respectively; the CSI for high thresholds (160–181 and 181–219) decreases even more sharply by 0.2984, with synthesis completely failing at the highest threshold range. These results indicate that lightning, as a sparse, yet high-value observation source, provides the model with crucial spatiotemporal cues, greatly enhancing its ability to identify and synthesize strong convective and extreme reflectivity structures.

Finally, by comparing 1 with 2 (resp. 6 with 7), it can be noticed that weighting the mse loss (resp. mse and SSIM loss functions) leads to an average improvement of 0.0285 (resp. 0.1349) in terms of the CSI. We believe that the included SSIM loss reflects the local contrast difference of the image and can measure the similarity of the structural features of the image. For radar data, high resolution and high density of information pose greater challenges to the model. As a result, the combined weighted loss function can guide the model to pay more attention to the local structure of different threshold intervals. The amount of high-threshold data is small, but the structure is obvious and the model pays more attention to it. Therefore, the synthesized radar image data at the junction of different threshold edges can be better predicted, and the overall CSI index has been improved.

In terms of visual results, by comparing Fig. 6(a)–(c) and (e)–(g), it can be observed that under unweighted conditions, the models using the MSFF block reconstruct more high-threshold regions (red patches). In addition, models using MSFF recover small structures more precisely, as shown in the serrated texture in Fig. 6(a) and (c).

Without using the MSFF, we compared Fig. 6(a) and (b) and Fig. 6(f) and (g). It can be seen that weighting has a significant impact on the reconstruction effect of high-threshold areas. Using weights to adjust the problem of data imbalance allows the model to perceive the same precision for both high and low thresholds. Comparing Fig. 6(d) and (e) and (i) and (j), it is noticeable that the synthesis accuracy of high-threshold regions significantly decreases when lightning data is not input. When using weights and MSFF, Fig. 6(e) and (j) demonstrates the impact of adding SSIM loss to VIL reconstruction. It can be observed that the overall structure in Fig. 6(j) is closer to the true VIL image, and there is no large-scale misgeneration in the low-threshold area. The reconstruction effect of the high-threshold area has also been improved.

2) Impact of the Window Size in the Swin Transformer: In this part, we analyze the impact of different window sizes, typically 4×4, 6×6, 8×8, and 12×12, on the performance of our SU3plus model.

As shown in Table V, the performance varies slightly across window sizes, reflecting a tradeoff between local detail modeling and global context aggregation. Overall, the best performance is achieved under the 6×6 and 8×8 settings. Specifically, the 6×6 setting achieves the highest average CSI (0.3862) and delivers the best performance at low thresholds (CSI16:0.6887 and CSI74:0.5220), indicating a well-balanced ability to capture both localized convective features and broader structural patterns.

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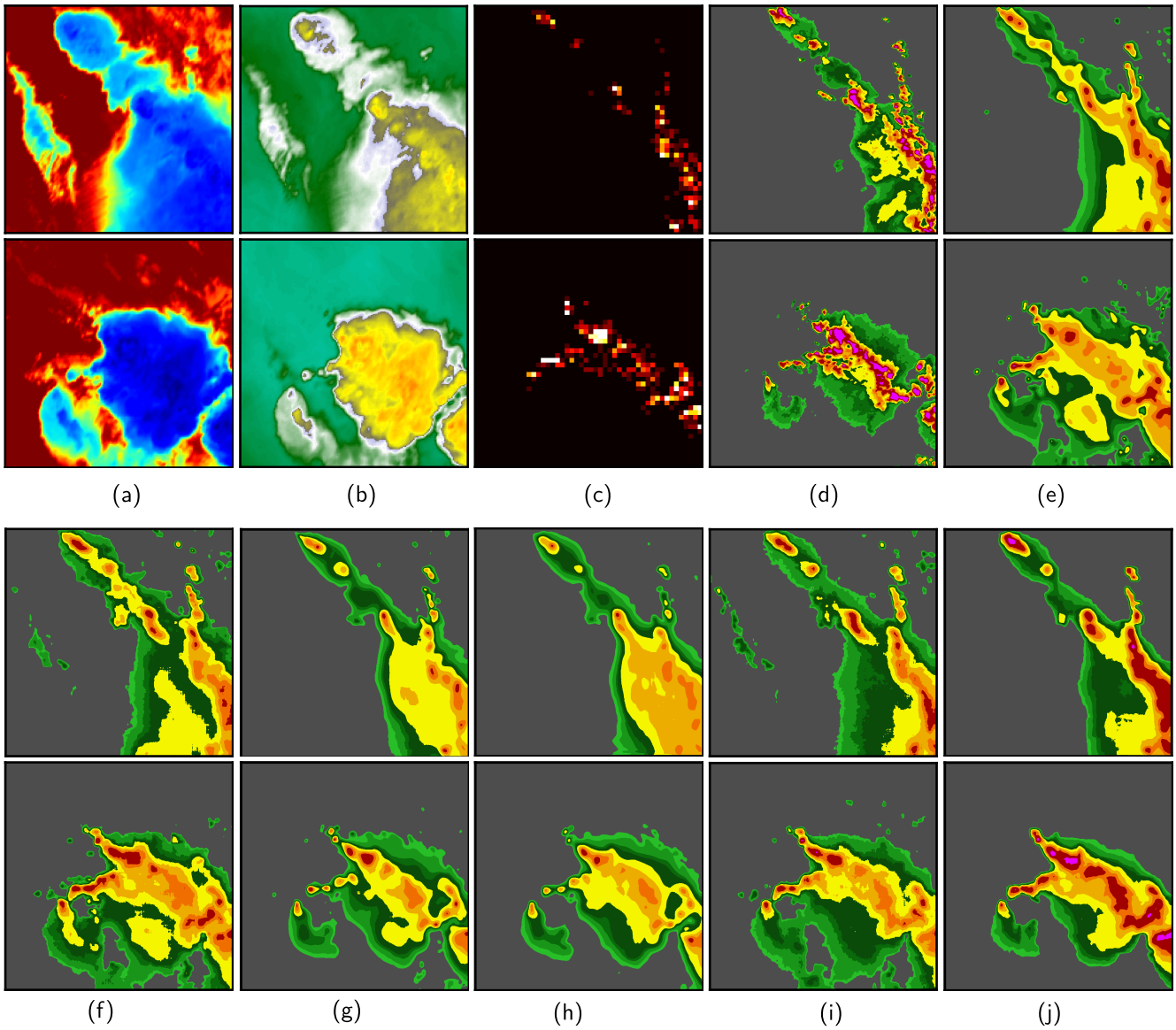


Fig. 5. Test data input images, real VIL images, and synthetic VIL images of each model. For the Swin Transformer, we added a convolutional layer at the last layer of the structure to match the output scale. As seen in the second-row images of the UNet [36] (e), Swin-UNet (f), and transUNet (i), these models reconstructed low-threshold regions that did not exist in the original VIL image. This region overlaps with cloud areas in IR-069 and IR-107 but does not appear in the real VIL image. It can be inferred that the physical information representing precipitation and other climate factors learned by these models is not accurate. For the high-threshold regions, our model (j) successfully reconstructed the pink areas representing high-threshold values, reflecting the true meteorological conditions. The other models performed poorly in reconstructing the pink and red regions. (a) IR-107. (b) IR-069. (c) LGHT. (d) TRUTH. (e) UNet [36]. (f) Swin-UNet [37]. (g) UNet3+ [38]. (h) Swin Transformer [45]. (i) TransUNet [39]. (j) SU3plus.

Moreover, from the visualization results shown in Fig. 7, the window of size 6×6 not only accurately restores the spatial distribution of strong echo regions, but also preserves the decaying structure at the boundaries. In contrast, the size of 4×4 does not fully reconstruct the main echo contours, while the size of 12×12 produces blurred and diffused centers, suggesting an inadequate representation of localized abrupt features.

It should also be noted that larger windows result in higher computational complexity. Therefore, a size window 6×6 is adopted in the design of our SU3plus model to achieve superior reconstruction of strong echo areas and better overall prediction accuracy.

3) Impact of Different Weighting Strategies in the Loss Function: Inspired by class imbalance mitigation via inverse frequency weighting [48], [49], [50], we introduce a loss weighting mechanism to improve the model's sensitivity to high-intensity regions. In radar images, such regions are critical, yet sparse; without weighting, the loss is dominated by background pixels, leading to biased learning. More precisely, we explore several weighting strategies.

1) *Raw:* It corresponds to the weight values, given by (7), by using the inverse frequency of each intensity bin.
2) *Rounded:* It represents the rounded version of raw values to reduce volatility and improve training stability.

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TABLE III

SEASONAL QUANTITATIVE EVALUATION OF SU3PLUS AND COMPARISON MODELS

Model / Season	SSIM↑	MSE↓	MAE↓	CSI16↑	CSI74↑	CSI133↑	CSI160↑	CSI181↑	CSI219↑	CSI-avg↑
UNet										
Spring	0.6603	993.65	14.60	0.5207	0.3912	0.2657	0.2780	0.2464	0.0871	0.2982
Summer	0.6251	1450.08	21.81	0.6221	0.4736	0.3325	0.3104	0.2715	0.0107	0.3368
Autumn	0.6450	1087.85	20.27	0.6250	0.3412	0.3250	0.3284	0.2859	0.0150	0.3201
Winter	0.4353	1525.94	23.53	0.5485	0.4165	0.1726	0.1613	0.1263	0.0215	0.2411
Att-UNet										
Spring	0.7113	942.68	13.63	0.5862	0.4275	0.2523	0.1962	0.0751	0.0071	0.2574
Summer	0.7231	1160.47	21.65	0.6742	0.5682	0.3064	0.2613	0.1130	0.0027	0.3210
Autumn	0.7448	807.62	13.46	0.6813	0.5311	0.3186	0.2574	0.0576	0.0064	0.3087
Winter	0.5053	1371.63	20.63	0.5673	0.4597	0.1584	0.1207	0.0098	0.0012	0.2195
Swin-UNet										
Spring	0.6560	987.53	15.84	0.5140	0.4263	0.3271	0.3315	0.2327	0.0037	0.3059
Summer	0.6307	1362.79	20.36	0.5362	0.4327	0.3587	0.3274	0.2162	0.0052	0.3127
Autumn	0.6571	1143.92	19.37	0.5271	0.4098	0.3473	0.3398	0.2301	0.0108	0.3108
Winter	0.4533	1473.62	22.85	0.4983	0.3824	0.2358	0.2510	0.1008	0.0003	0.2448
UNet3+										
Spring	0.7460	728.47	11.84	0.6190	0.4220	0.2870	0.1783	0.1317	0.0020	0.2733
Summer	0.7105	923.58	15.47	0.6984	0.5362	0.3369	0.2163	0.1545	0.0035	0.3243
Autumn	0.7320	730.46	19.30	0.6873	0.5517	0.3471	0.2207	0.1292	0.0042	0.3234
Winter	0.5180	1195.13	21.45	0.6207	0.4903	0.2362	0.1462	0.0857	0.0007	0.2633
Swin-Transformer										
Spring	0.7413	765.34	12.91	0.5820	0.4348	0.2762	0.1858	0.0262	0.0001	0.2508
Summer	0.7098	960.83	15.86	0.6881	0.5653	0.3398	0.2576	0.0324	0.0001	0.2390
Autumn	0.7311	850.02	18.24	0.6894	0.5237	0.3421	0.2483	0.0308	0.0002	0.3058
Winter	0.5162	1245.56	21.57	0.6501	0.5106	0.2143	0.1330	0.0154	0	0.2539
TransUNet										
Spring	0.7150	785.62	13.63	0.5651	0.4251	0.2944	0.3081	0.2554	0.0093	0.3096
Summer	0.6880	1010.66	17.15	0.6380	0.5012	0.3456	0.3351	0.2905	0.0118	0.3537
Autumn	0.7050	880.73	19.11	0.6431	0.4652	0.3367	0.3323	0.2851	0.0132	0.3459
Winter	0.4820	1275.98	22.18	0.5834	0.4351	0.1945	0.1821	0.1411	0.0031	0.2566
SU3plus (Ours)										
Spring	0.7680	599.89	10.95	0.6038	0.4473	0.3114	0.3226	0.2938	0.1557	0.3558
Summer	0.7271	831.05	14.39	0.6906	0.5643	0.3880	0.3816	0.3462	0.1613	0.4220
Autumn	0.7621	660.79	18.83	0.6958	0.5334	0.3830	0.3867	0.3510	0.1817	0.4219
Winter	0.5631	1095.07	19.43	0.6410	0.5084	0.2630	0.2643	0.2180	0.0762	0.3285

TABLE IV

EVALUATION SCORES OF DIFFERENT MODELS IN THE ABLATION STUDY

Loss	#	MSFF	Weight	LGHT	CSI16	CSI74	CSI133	CSI160	CSI181	CSI219	CSI-avg
MSE	1	✗	✗	✓	**0.6635**	**0.4963**	0.2558	0.1694	0.0630	4E-04	0.2747
	2	✗	✓	✓	0.5081	0.4162	0.3379	0.3346	**0.2178**	0.0048	0.3032
	3	✓	✗	✓	0.6625	0.4944	0.2665	0.1836	0.0777	9E-04	0.2809
	4	✓	✓	✗	0.6619	0.4124	0.1117	0.0504	0.0133	0	0.2083
	5	✓	✓	✓	0.5505	0.4700	**0.3443**	**0.3618**	0.2019	**0.0078**	**0.3152**
MSE + SSIM	6	✗	✗	✓	0.6773	0.4638	0.1681	0.0695	0.0112	5E-06	0.2317
	7	✗	✓	✓	0.6729	0.5092	0.3541	0.3462	0.2772	0.0400	0.3666
	8	✓	✗	✓	0.6707	0.4853	0.2266	0.1225	0.0284	5E-05	0.2556
	9	✓	✓	✗	0.6557	0.4382	0.1200	0.0482	0.0075	0	0.2113
	10	✓	✓	✓	**0.6887**	**0.5220**	**0.3551**	**0.3502**	**0.3059**	**0.0950**	**0.3862**

TABLE V

PERFORMANCE ANALYSIS OF OUR SU3PLUS MODEL UNDER DIFFERENT WINDOW SIZES

Window Size	SSIM↑	MSE↓	MAE↓	CSI16↑	CSI74↑	CSI133↑	CSI160↑	CSI181↑	CSI219↑	CSI-avg↑
12×12	0.7530	736.8498	12.3857	0.6657	0.5054	0.3572	0.3546	0.3070	0.0979	0.3813
8×8	**0.7566**	**701.4137**	**12.1268**	0.6757	0.5109	**0.3655**	**0.3597**	**0.3117**	0.0909	0.3857
6×6	0.7545	711.7410	12.2806	**0.6887**	**0.5220**	0.3551	0.3502	0.3059	0.0950	**0.3862**
4×4	0.7472	787.8940	12.7798	0.6691	0.5017	0.3474	0.3484	0.3105	**0.1232**	0.3834

3) *Log-Scaled:* It is the logarithmic transformation of raw values to compress their range.
4) *Capped @T:* It applies a maximum cap given by a threshold T (e.g., $T \in \{50, 100, 200\}$) on raw values to avoid extreme weights while retaining contrast.
5) *No Weight:* It represents the baseline loss without any weighting strategy.

The resulting weight values across the different intensity bins are shown in Table VI. Moreover, the performance of SU3plus, obtained with these weighting strategies, is depicted

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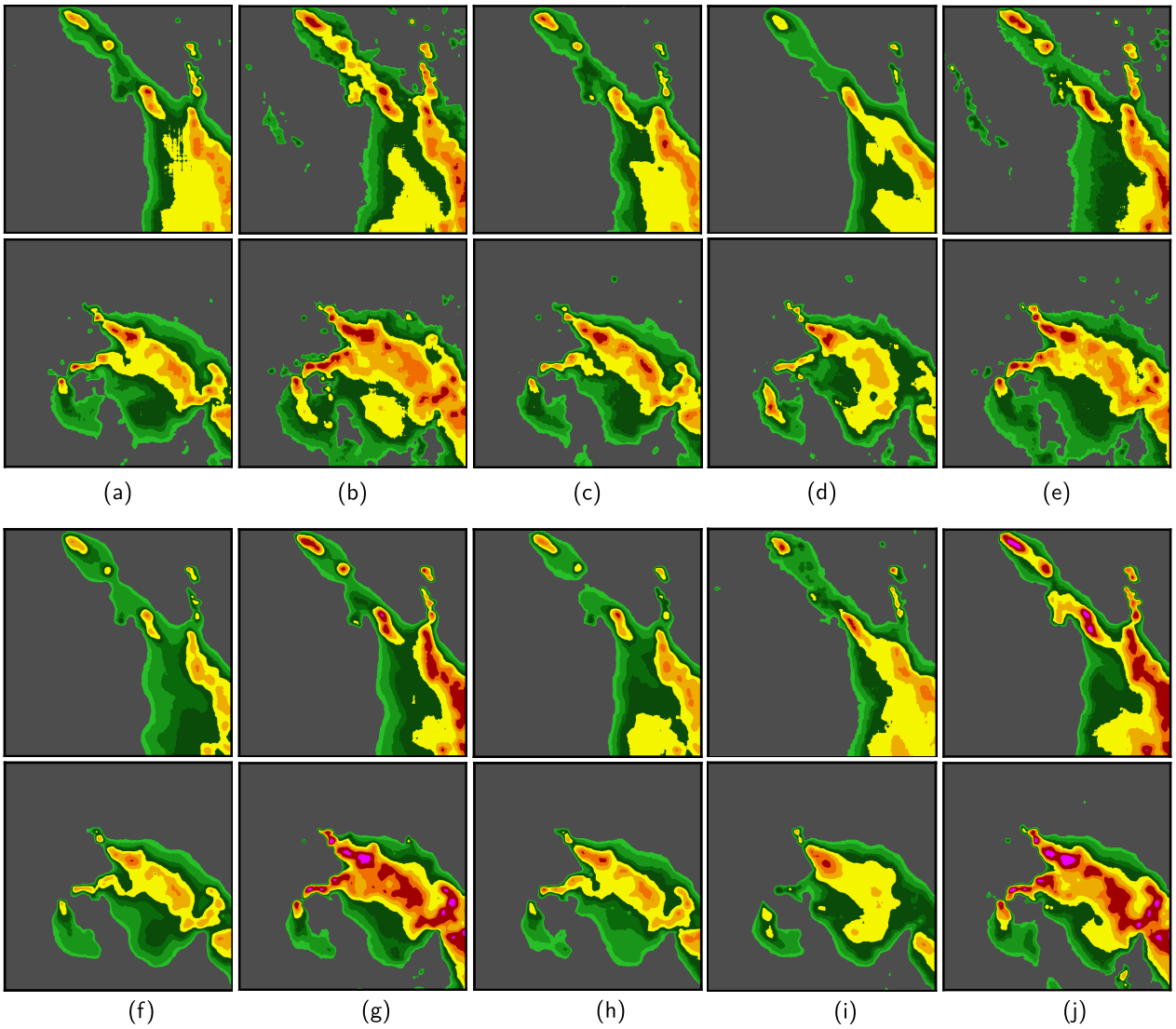


Fig. 6. Image visualization of different ablation experiment models. By comparing (a)–(c) and (f)–(h), it can be seen that under unweighted conditions, models using the MSFFB reconstructed more high-threshold areas (red patches). Using weights to adjust the problem of data imbalance allows the model to perceive the same precision for both high and low thresholds. Comparing (d) and (e) and (i) and (j), it is noticeable that the synthesis accuracy of high-threshold regions significantly decreases when lightning data is not input. When using weights and the MSFF block, we compared (e)–(j) to reflect the effect of adding the SSIM loss on VIL reconstruction. It can be observed that the overall structure in (j) is closer to the real VIL image, and there is no large-scale misgeneration in the low-threshold area. (a) #1. (b) #2. (c) #3. (d) #4. (e) #5. (f) #6. (g) #7. (h) #8. (i) #9. (j) #10.

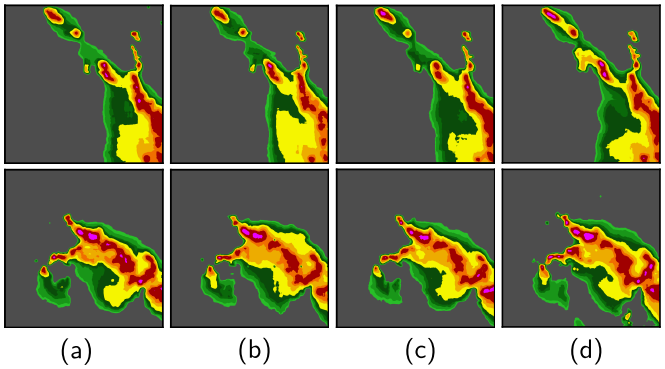


Fig. 7. Image visualization of our SU3plus model under different window sizes. (a) 12 × 12. (b) 8 × 8. (c) 6 × 6. (d) 4 × 4.

in Table VII. Thus, the unweighted model almost completely fails in high-intensity events (e.g., CSI181 = 0.0284 and CSI219 = 0.00005), confirming the necessity of the weighting

TABLE VI
WEIGHT VALUES ACROSS INTENSITY BINS

Weight Type	0-16	16-74	74-133	133-160	160-181	181-219	> 219
Raw	1.3363	8.4795	10.7041	45.0041	117.9627	133.3007	476.5719
Rounded	1	8	10	50	120	140	480
Log-scaled	0.290	2.139	2.369	3.807	4.770	4.892	6.166
Capped @50	1	8	10	45	50	50	50
Capped @100	1	8	10	45	100	100	100
Capped @200	1	8	10	45	120	140	200

procedure. Most importantly, the retained weighting strategy (i.e., *raw*), *rounded*, and *capped @100* yield similar reconstruction quality and achieve the highest CSI average, while maintaining good performance at higher thresholds.

F. Computational Complexity

The proposed SU3plus model has also been compared with two main baseline models, Including the UNet and

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TABLE VII
PERFORMANCE OF THE SU3PLUS MODEL UNDER DIFFERENT LOSS WEIGHTING STRATEGIES

Weight Type	SSIM↑	MSE↓	MAE↓	CSI16↑	CSI74↑	CSI133↑	CSI160↑	CSI181↑	CSI219↑	CSI-avg↑
Raw	0.7547	753.3624	12.7275	0.6759	0.5065	0.3576	**0.3587**	**0.3166**	0.1007	0.3860
No Weight	0.7095	928.7056	19.2107	0.6707	0.4853	0.2266	0.1225	0.0284	0.00005	0.2556
Rounded	0.7557	725.4558	12.2657	0.6711	0.5079	0.3562	0.3600	0.3143	**0.1017**	0.3852
Log-scaled	0.7545	**594.7357**	**10.9167**	0.6726	0.4951	0.2446	0.1854	0.1006	0.0112	0.2849
Capped @50	**0.7598**	663.3604	11.9820	**0.6899**	0.5133	**0.3598**	0.3282	0.2305	0.0158	0.3563
Capped @100	0.7545	711.7410	12.2806	0.6887	**0.5220**	0.3551	0.3502	0.3059	0.0950	**0.3862**
Capped @200	0.7559	726.1753	12.3608	0.6720	0.5065	0.3543	0.3564	0.3093	0.0671	0.3776

TABLE VIII
COMPUTATIONAL COMPLEXITY EVALUATION FOR SU3PLUS AND BASELINE MODELS

Model	Inference Time (s)	Memory Usage (MB)	Param. (M)	FLOPs (G)
SU3plus	0.2134	3910	34.6	18.5
U-Net	0.0388	8510	17.5	34.57
Swin-Unet	0.2031	3408	26.36	16.62

Swin-Unet, in terms of inference time, memory usage, parameters, and FLOPs. This comparison is illustrated in Table VIII. Overall, the Transformer-based architectures SU3plus and Swin-Unet show very similar inference time, memory usage, and FLOPs. Moreover, SU3plus introduces additional computational complexity compared to the UNet, but achieves superior reconstruction performance.

It should also be noted that our model is designed for use with geostationary meteorological satellites (e.g., GOES-16), which typically provide observations every 10–15 min, and at a minimum of 1-min intervals under rapid-scan mode [51]. Therefore, any model capable of achieving inference within 1 min satisfies the operational real-time requirements. SU3plus requires an average inference time of 0.2134 s per frame, showing great potential for deployment in near-real-time systems.

G. Discussion

This study aims to develop a DL model based on the Transformer, using geosynchronous meteorological satellite observation data to reconstruct radar composite reflectivity data. In this respect, we use the SEVIR storm dataset for training and testing and transform IR-107, IR-069, and LGHT data into radar composite reflectivity data. By designing the SU3plus network, we have confirmed that the Swin Transformer module has stronger feature extraction capabilities for radar composite reflectivity data than convolutional modules. This conclusion is consistent with our conjecture that the Transformer architecture is more suitable for application to Earth system observation data. In response to the characteristic of radar data having more spatial scale detail features, we design the SU3plus network with an MSFF block, while leveraging lightning information. Compared to traditional CNN- or RNN-based approaches, SU3plus exhibits significant advantages in spatiotemporal structure modeling. By incorporating a hierarchical Swin Transformer module, it leverages window-based self-attention and shift mechanisms to overcome the limitations of fixed receptive fields, enabling effective modeling of long-range spatial dependencies. Moreover, it can automatically focus on critical regions such as strong convection areas, enhancing the reconstruction of localized extreme structures. Furthermore, SU3plus's multiscale fusion design is better suited to the wide range of cloud system scales present in meteorological imagery, providing methodological support for the fine-grained modeling of complex weather systems.

In addition to the design of network architecture, we also optimized the learning strategy. First, according to the characteristic of rich edge details in radar images, we included an SSIM loss to improve the restoration of details. Furthermore, we analyzed the probability statistics on the training set of the SEVIR storm dataset and found that the threshold interval distribution in the data is unbalanced. This characteristic is also consistent with the real weather conditions in reality. However, this characteristic of radar data makes the synthesis of high threshold intervals difficult. To address this problem, we defined a weighted loss function to improve the model's performance. This loss combines a weighted mse loss and a weighted SSIM loss. The resulting SU3plus model has achieved further improvement and was found to be the most effective in terms of CSI index and quality of reconstruction.

V. CONCLUSION

This article proposes a network for the reconstruction of radar VIL. The network architecture is mainly based on the Transformer model, which is suitable for capturing local and long-distance dependencies existing in meteorological satellite and radar data. In this respect, we designed an MSFF block to guide our model in paying attention to the high-threshold area and overall structure of the data and defined a new loss function combining the weighted SSIM and mse metrics. Extensive experiments, carried out on the SEVIR dataset, are performed to compare our model with several state-of-the-art models. The results show that the proposed methodology achieves the best results. Although our model can better pay attention to high-threshold areas and details, there is still a lack of balance between low-threshold and high-threshold effects. In future work, we aim to employ diffusion models to reconstruct VIL data, improving both its accuracy and reliability. Incorporating radiative transfer models or leveraging radar reflectivity equations to impose physical constraints could further enhance the interpretability and generalizability of the proposed model. Furthermore, generative large language

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models can be integrated to provide semantic guidance for VIL synthesis using multimodal information.

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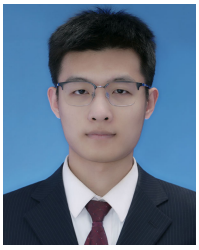
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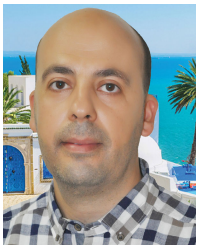
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